



Data Communications and Networking

Fourth Edition

Chapter 1

Introduction



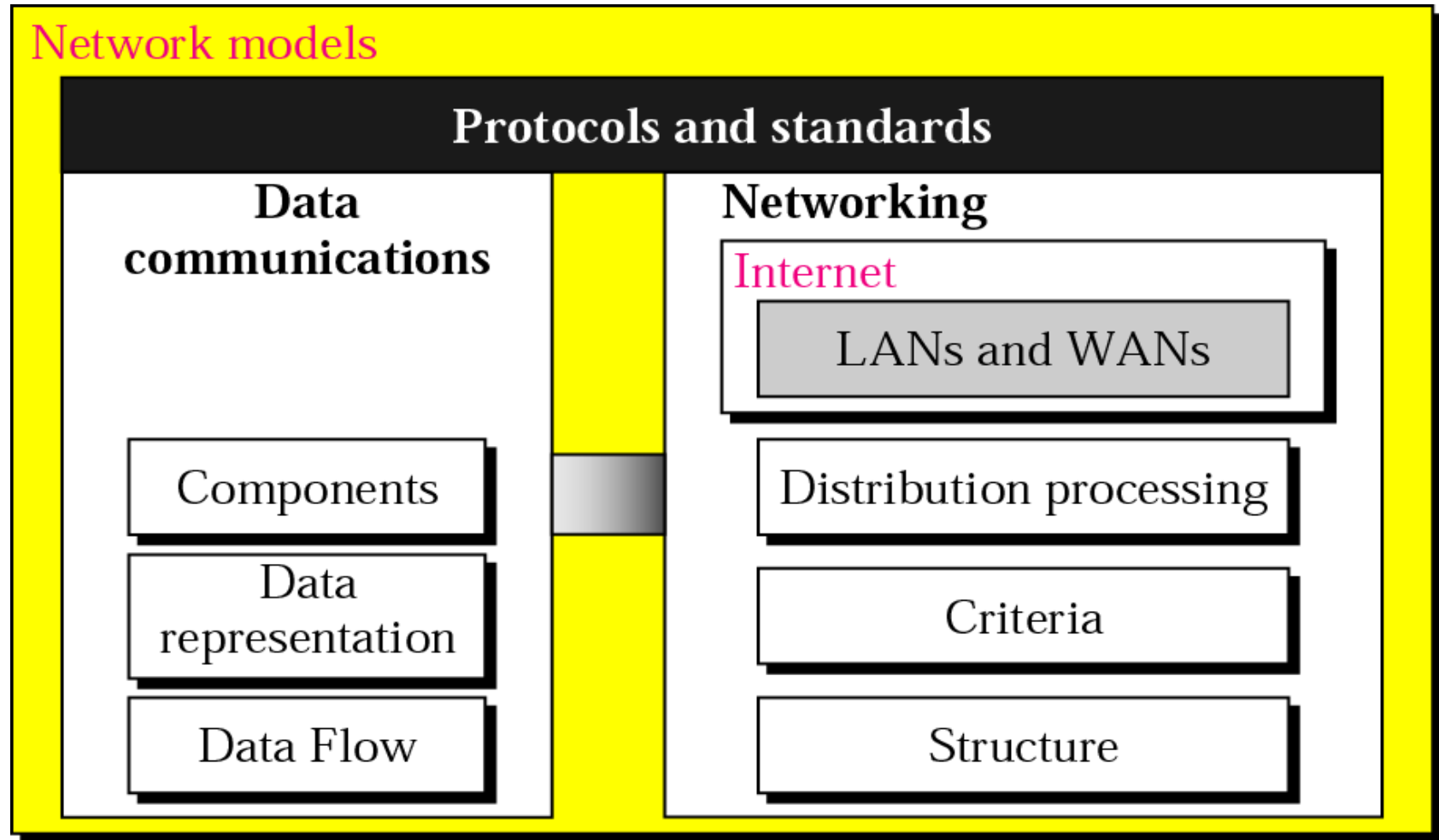
Presentation / Lecture Acknowledgement

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Data Communications and Networking

Network models



1.1 DATA COMMUNICATIONS

The term **telecommunication** means communication at a distance. The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data. **Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Topics discussed in this section:

Components

Data Representation

Data Flow



Fundamental Characteristics

❑ The effectiveness of a data communication system depends on three fundamental characteristics

❖ Delivery

- must deliver data to the correct destination
- (the intended device or user)

❖ Accuracy

- must deliver data accurately

❖ Timeliness

- must deliver data in a timely manner

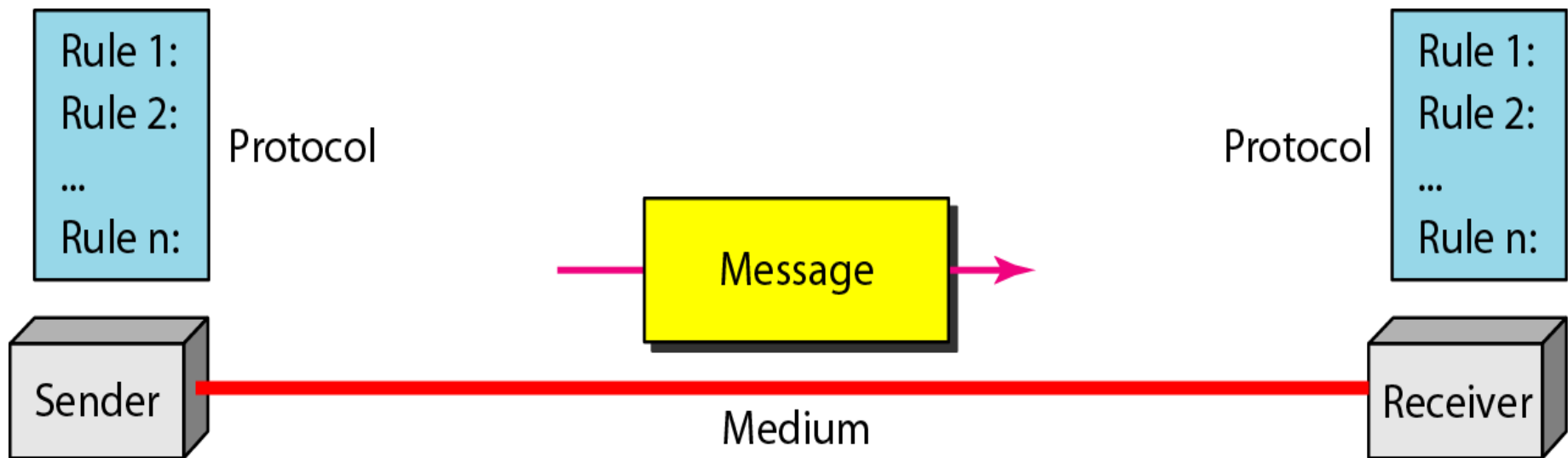
❖ Jitter

- Refers to the variation in the packet arrival time.



Data communication system components

Figure 1.1 *Five components of data communication*



Data communication system components (cont'd)

❑ Message

❖ is the information(data) to be communicated.

(text, numbers, pictures, sound, video or any combination of these)

❑ Sender

❖ is the device that sends the data message.

(computer, workstation, telephone handset, video camera)

❑ Receiver

❖ is the device that receives the message.

(computer, workstation, telephone handset, video camera)



Data communication system components (cont'd)

❑ Medium

- ❖ is the physically path by which a message travels from sender to receiver.

(twisted pair wire, coaxial cable, fiber- optic cable, laser , or radio waves(terrestrial or satellite microwave))

❑ Protocol

- ❖ is a set of rules that govern data communication.



Data Presentation

❑ Text is represented as a bit pattern.

- ◆ Code : Each sets of bit patterns to represent text symbols.
- ◆ ASCII : by ANSI, 7bits
- ◆ Extended ASCII, 8bits
- ◆ Unicode : 16 bits
- ◆ ISO : 32bits

❑ Numbers : also represented by bit patterns.

❑ Images : also represented by bit patterns. (Picture)

❑ Audio : It is continuous, not discrete. (Voice or music)

❑ Video : The recording or broadcasting of a picture or movie.



Direction of Data Flow

- ❖ is used to define the direction of signal flow between two link devices

□ Simplex

- ◆ unidirectional, as on a one-way street (keyboard, monitor)

□ Half-Duplex

- ◆ each station can both transmit and receive, but not at the same time

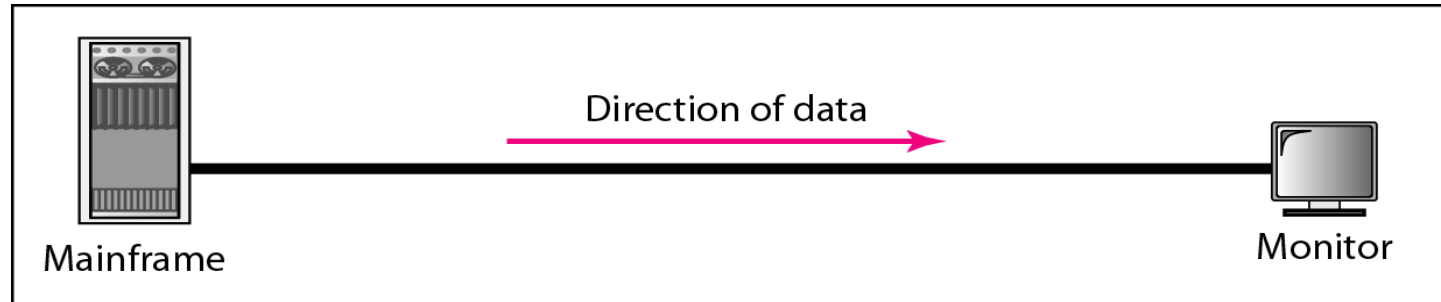
□ Full-Duplex

- ◆ both stations can transmit and receive simultaneously

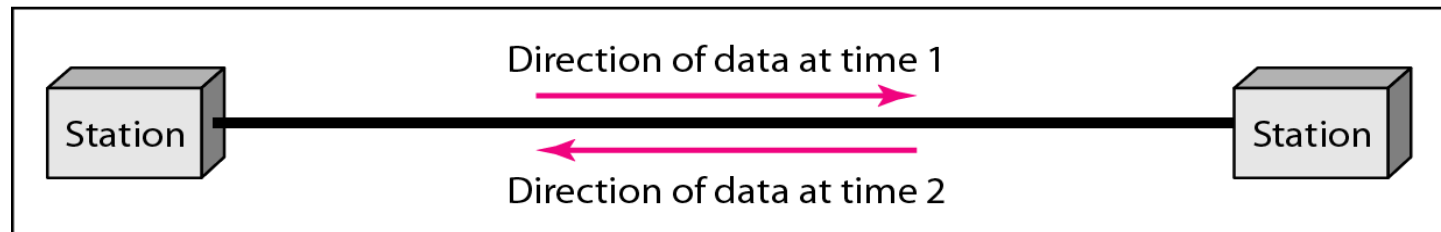


Direction of Data Flow (cont'd)

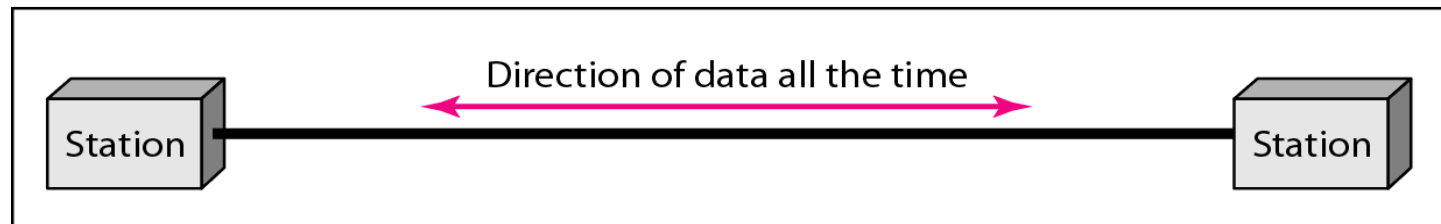
Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



a. Simplex



b. Half-duplex



c. Full-duplex

1.2 NETWORKS

A **network** is a set of devices (often referred to as **nodes**) connected by communication **links**. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

Topics discussed in this section:

Distributed Processing

Network Criteria

Physical Structures

Network Models

Categories of Networks

Interconnection of Networks: Internetwork

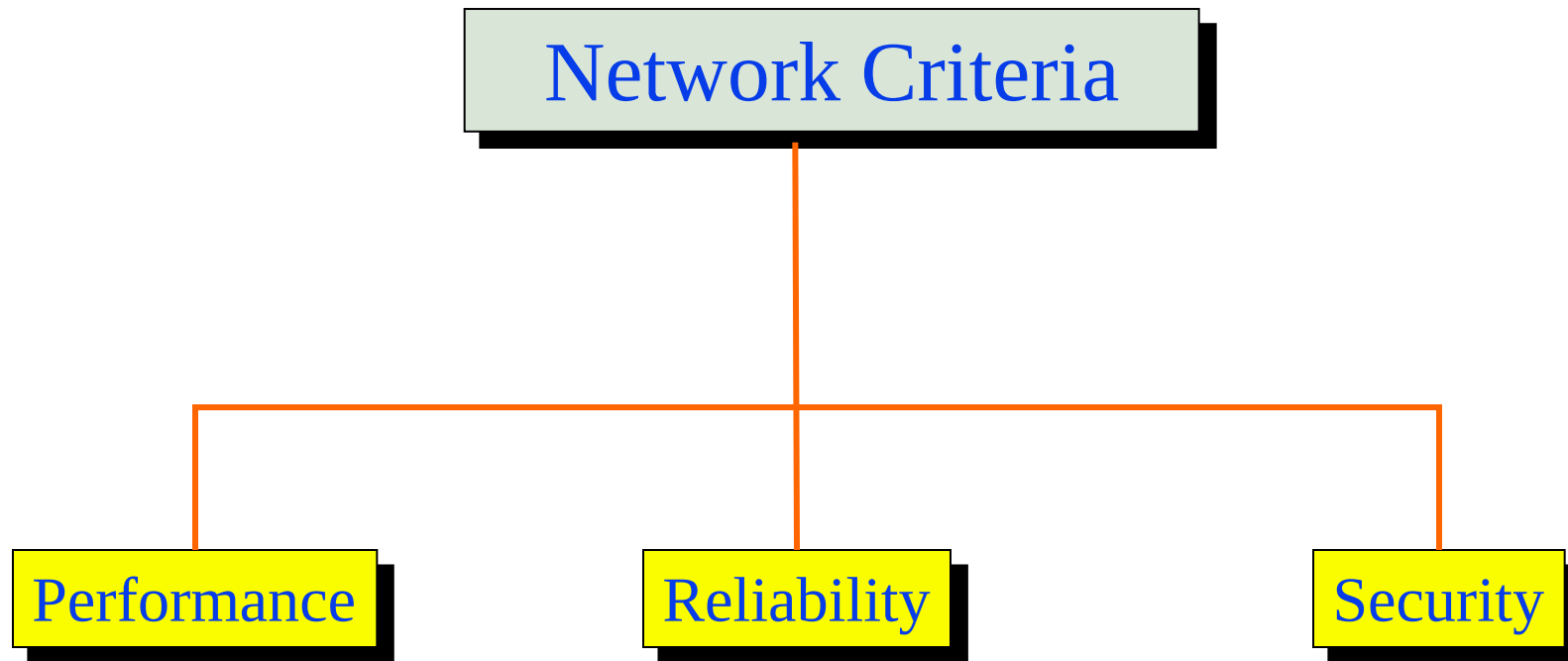


Distributed processing

- ❑ A task is divided among multiple computers.
- ❑ Instead of one single large machine being responsible for all aspects of a process, separate computers handle a subset.



Network Criteria



Performance

□ Depends on a number of factors.

- ◆ number of users
- ◆ type of transmission medium(data rate)
- ◆ hardware
- ◆ software : error-free delivery

□ Evaluated by two networking metrics:

◆ Throughput and Delay.



Reliability

- ❑ Frequency of failure
- ❑ Recovery time of a network after a failure
- ❑ Catastrophe
 - ❖ Networks must be protected from catastrophic events such as fire, earthquake, or theft.



Security

- ❑ **Protecting data from Unauthorized access**
- ❑ **Protecting data from Viruses and Worms**



Physical Structures

❑ Type of Network Connection

❖ Point-to-point

- provides a dedicated link between two devices.

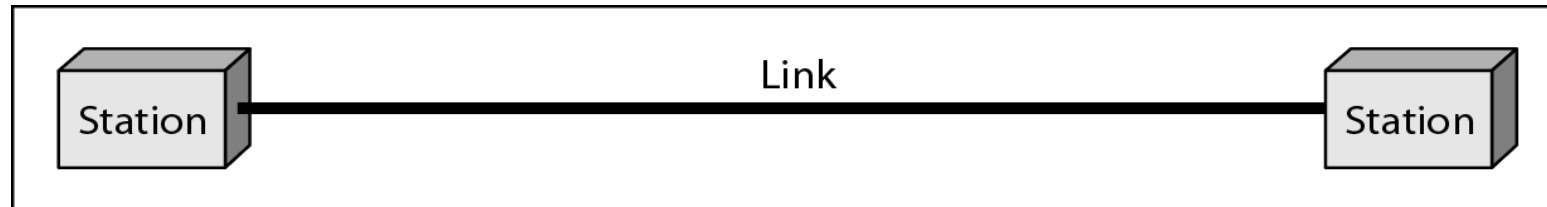
❖ Multipoint : multi-drop

- is configuration in which more than two specific devices share a single link

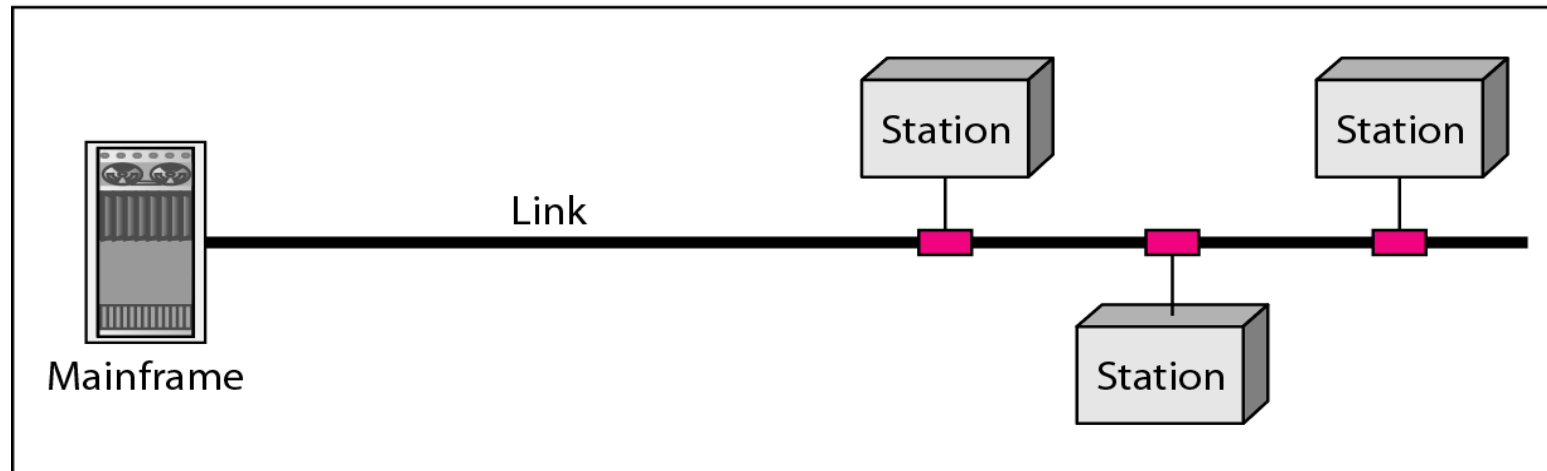


Physical Structures (cont'd)

Figure 1.3 *Types of connections: point-to-point and multipoint*

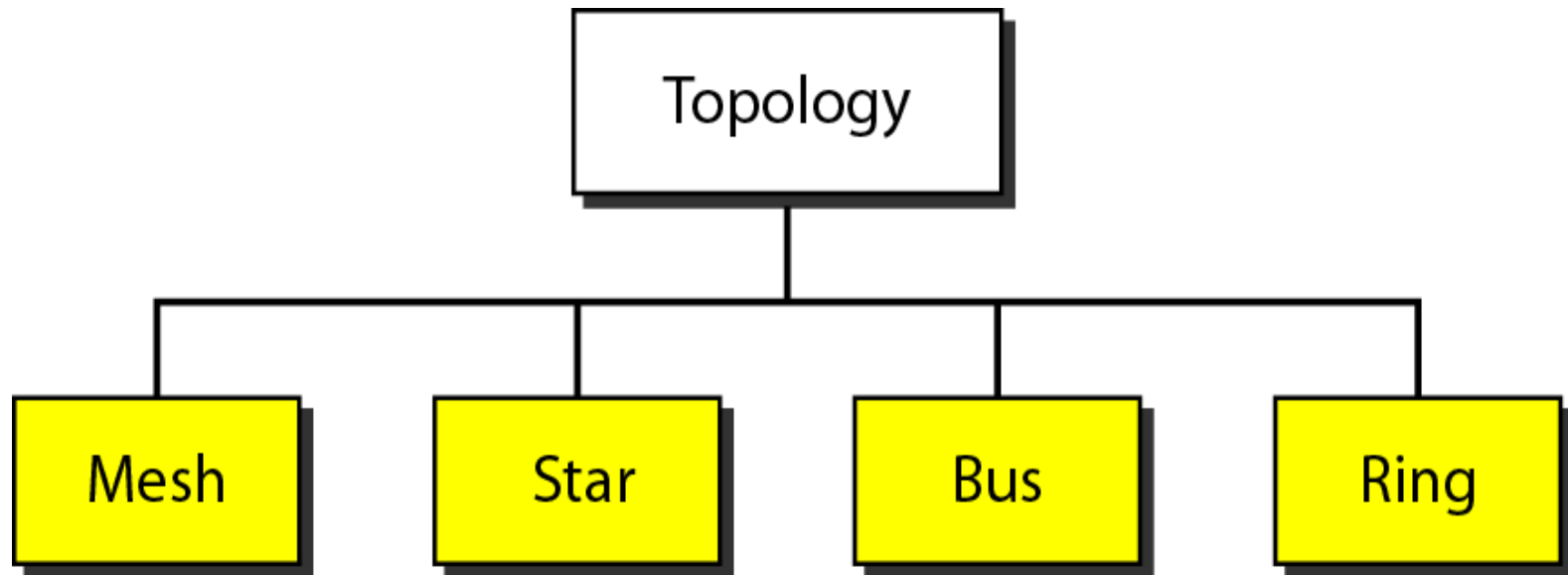


a. Point-to-point



b. Multipoint

Physical Topology



Physical Topology

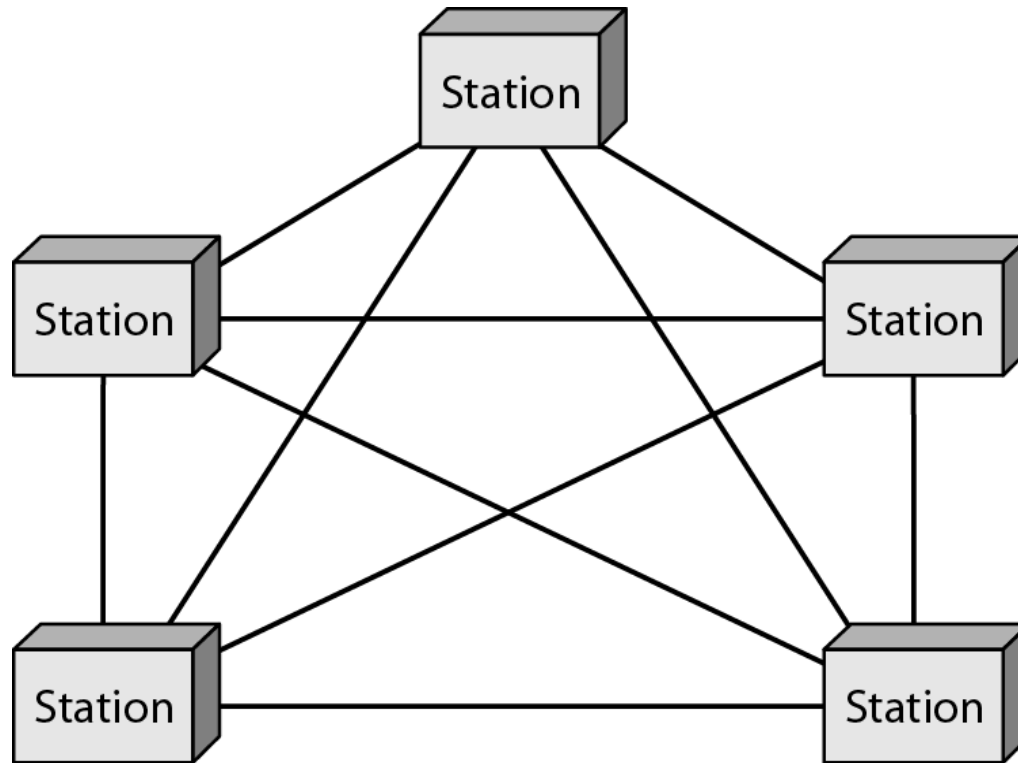
□ Mesh

- ❖ Every device has a dedicated point-to-point link to every other device.
- ❖ A fully connected mesh network therefore has $n(n-1)/2$ physical channels to link n devices.



Mesh

Mesh



Mesh (cont'd)

□ Advantages

- ◆ The use of dedicated links guarantees that each connection can carry its data load.
- ◆ Mesh topology is robust.
- ◆ Privacy and security.
- ◆ Point-to-point links make fault identification and fault isolation easy.



Mesh (cont'd)

❏ Disadvantages

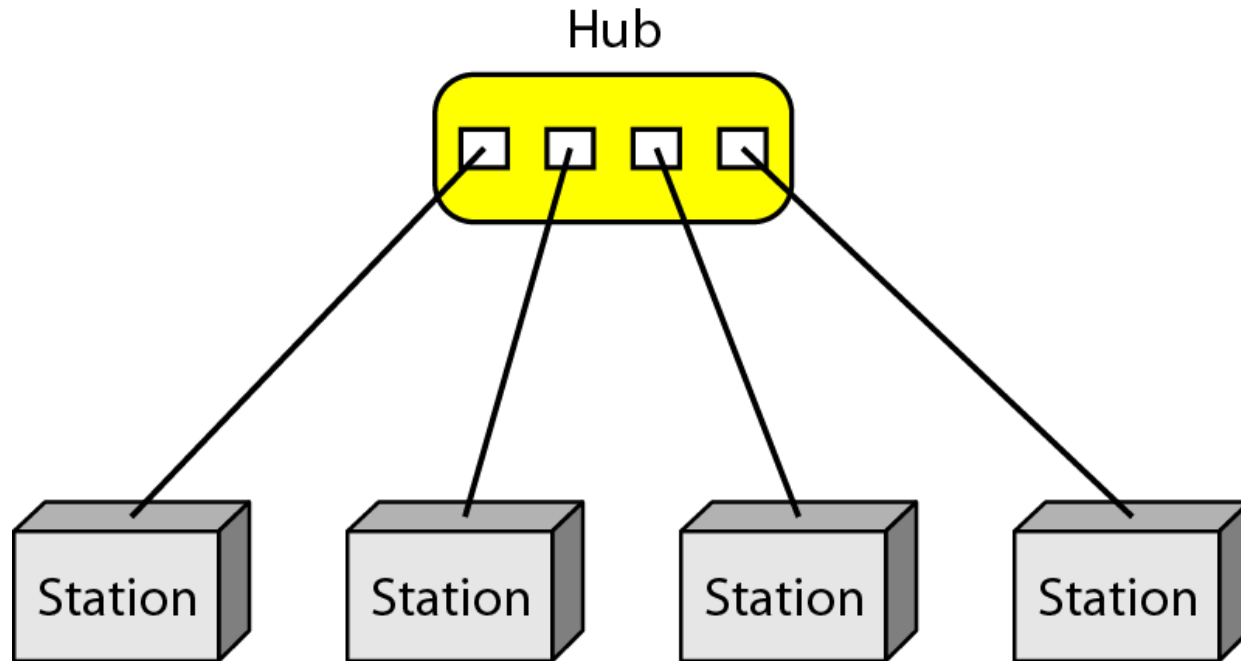
- ◆ are related to the amount of cabling and the number of I/O ports
- ❖ because every device must be connected to every other device, installation and reconfiguration are difficult
- ❖ the sheer bulk of the wiring can be greater than the available space(in walls, ceiling, or floors) can accommodate
- ❖ the hardware required to connect each link (I/O port and cable) can be prohibitively expensive



Star Topology

- ❑ Each device has a dedicated point-to-point link only to a central controller, usually called a hub

- ❑ Star Topology



Star (cont'd)

❏ Advantage

- ❖ Each device needs only one link and one I/O port to connect it to any number of others (easy to install and reconfigure)
- ❖ Robustness
 - if one link fails, only that link is affected



Tree Topology

❑ Tree topology is a variation of a star

❖ active hub(central hub)

- contains a repeater, which is a hardware device that generates the received bit patterns before sending them out

❖ passive hub

- provides a simple physical connection between the attached devices

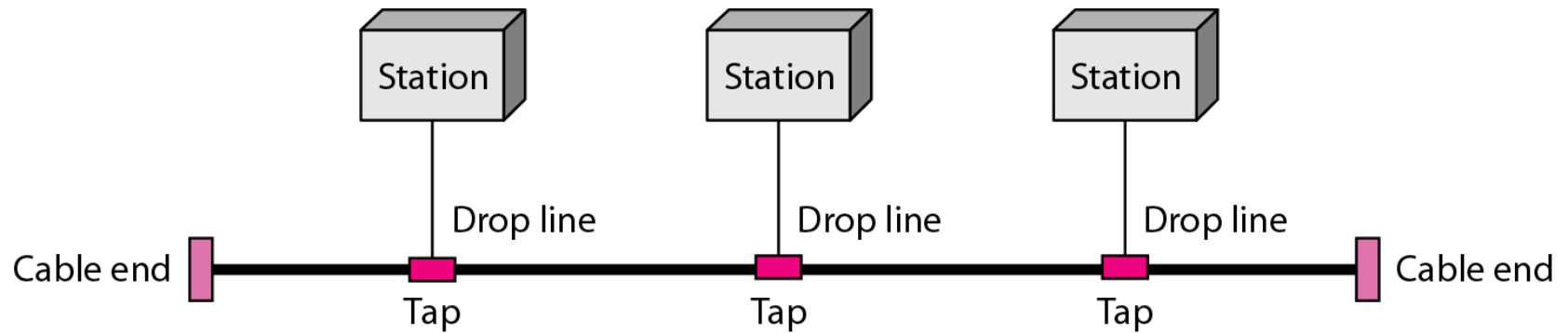


Bus Topology

- ❑ Bus topology is multipoint. One long cable acts as a backbone to link all the devices in the network
- ❑ Nodes are connected to the bus cable by drop lines and taps
 - ❖ Drop line
 - is a connection running between the device and the main cable
 - ❖ Tap
 - is a connector either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core



Bus Topology (cont'd)



Bus (cont'd)

❑ Advantages

- ◆ include ease of installation

❑ Disadvantages

- ◆ include difficult reconfiguration and fault isolation



Ring Topology

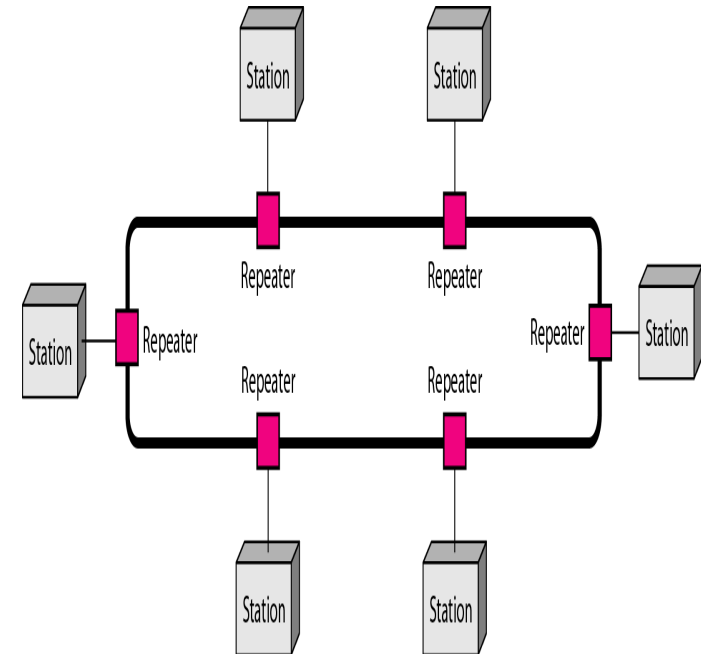
❑ Each device has a dedicated point-to-point line configuration only with the two devices on either side of it

❑ Advantage

- ◆ is relatively easy to install and reconfigure
- ◆ fault isolation is simplified

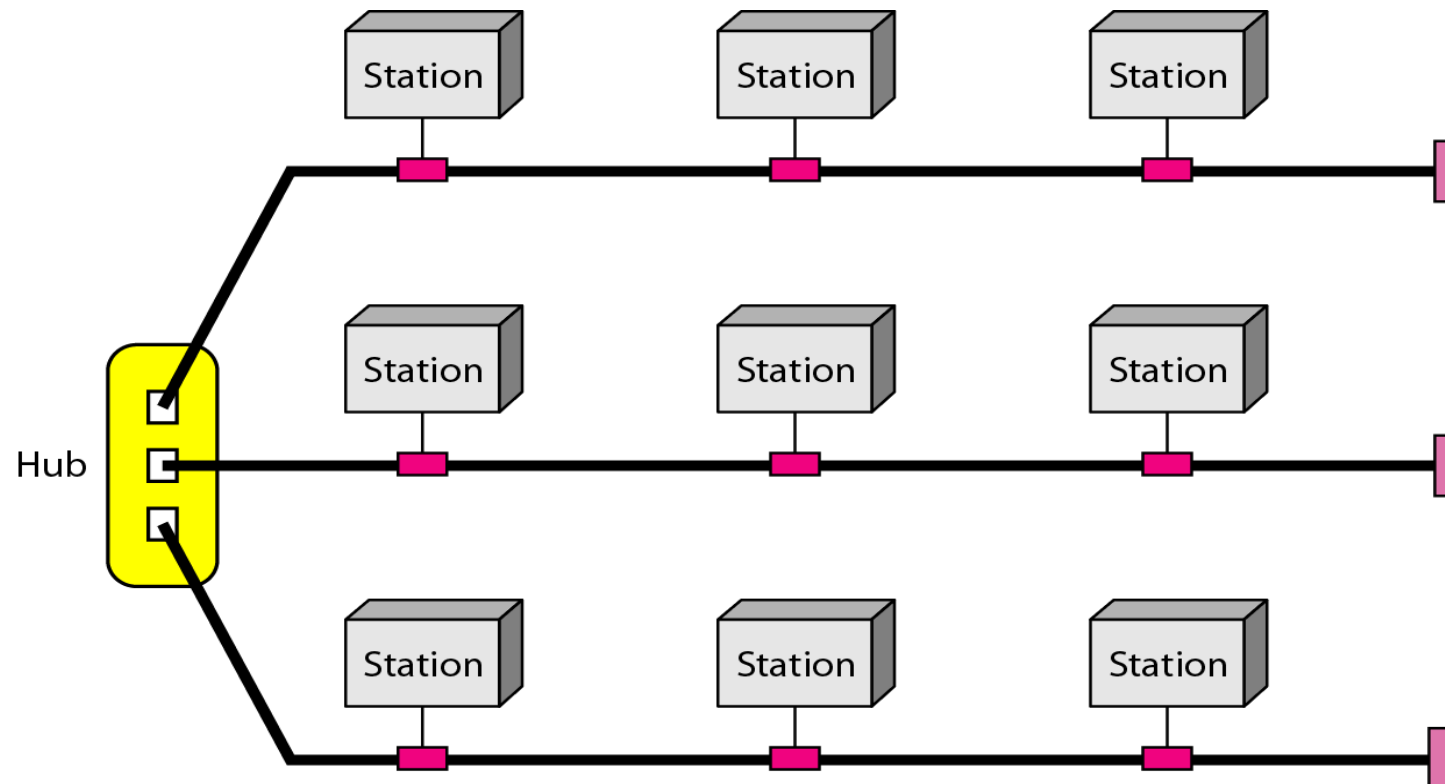
❑ Disadvantage

- ❖ unidirectional traffic
 - break in the ring can disable the entire network
 - > needs dual ring



Hybrid Topology

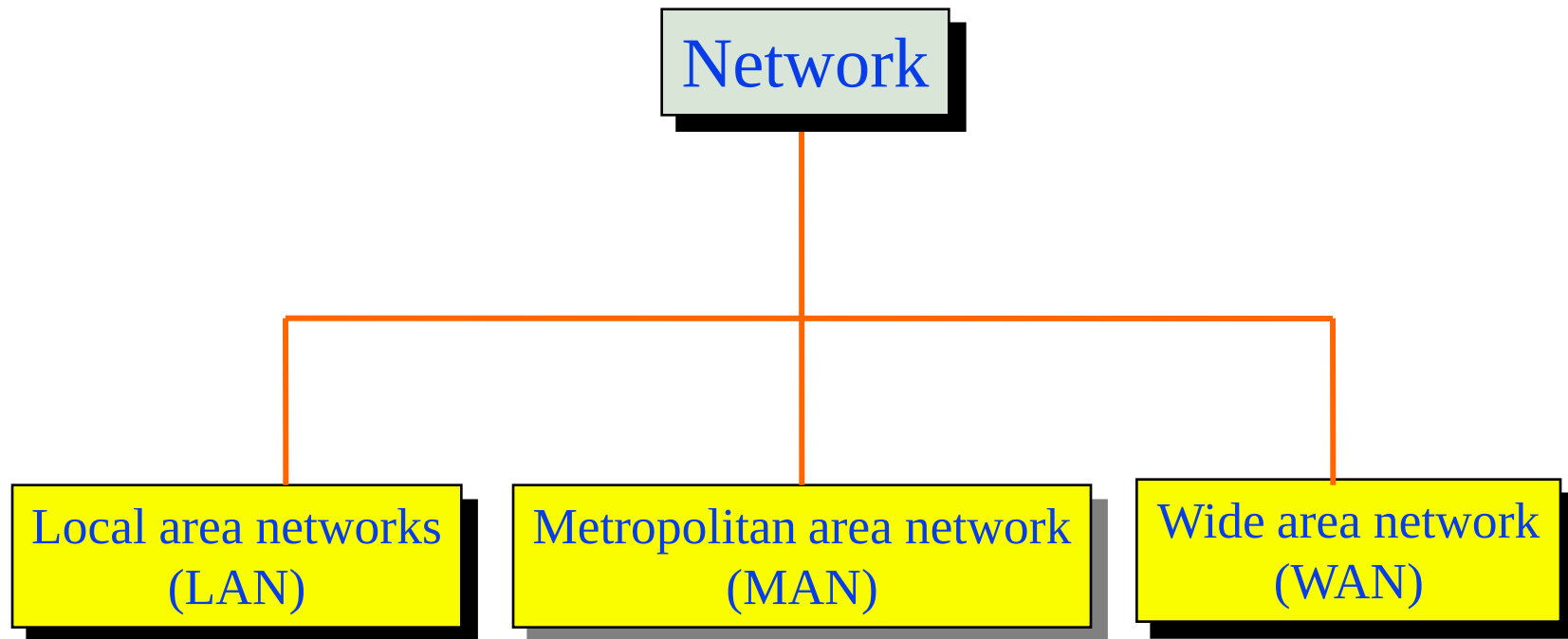
- ❑ A network can be hybrid.
- ❑ For example, a main star topology with each branch connecting several stations in a bus topology.



Categories of Networks

❑ Three primary categories

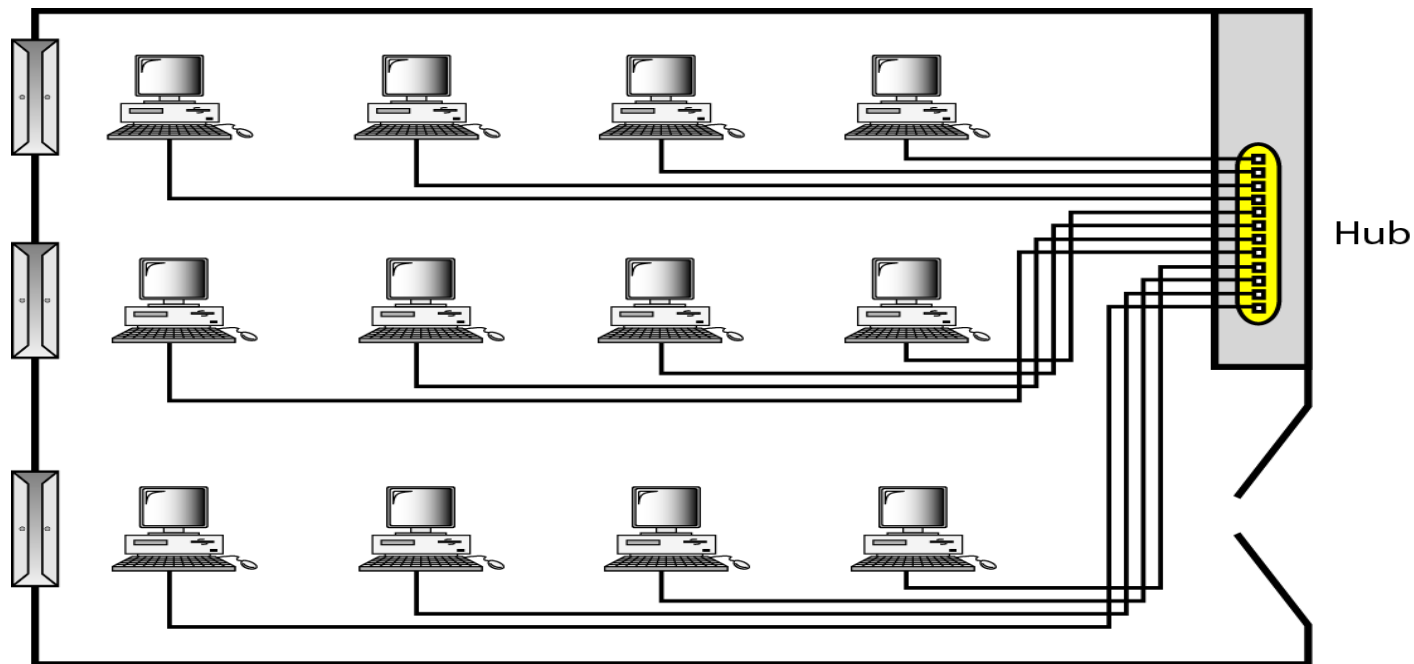
❖ size, ownership, distance it cover, physical architecture



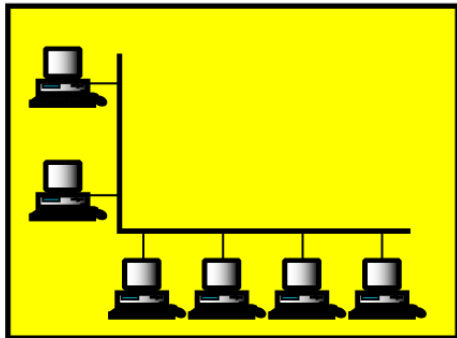
LAN (Local Area Networks)

- ❑ LAN is usually privately owned and links the devices in a single office, building or campus

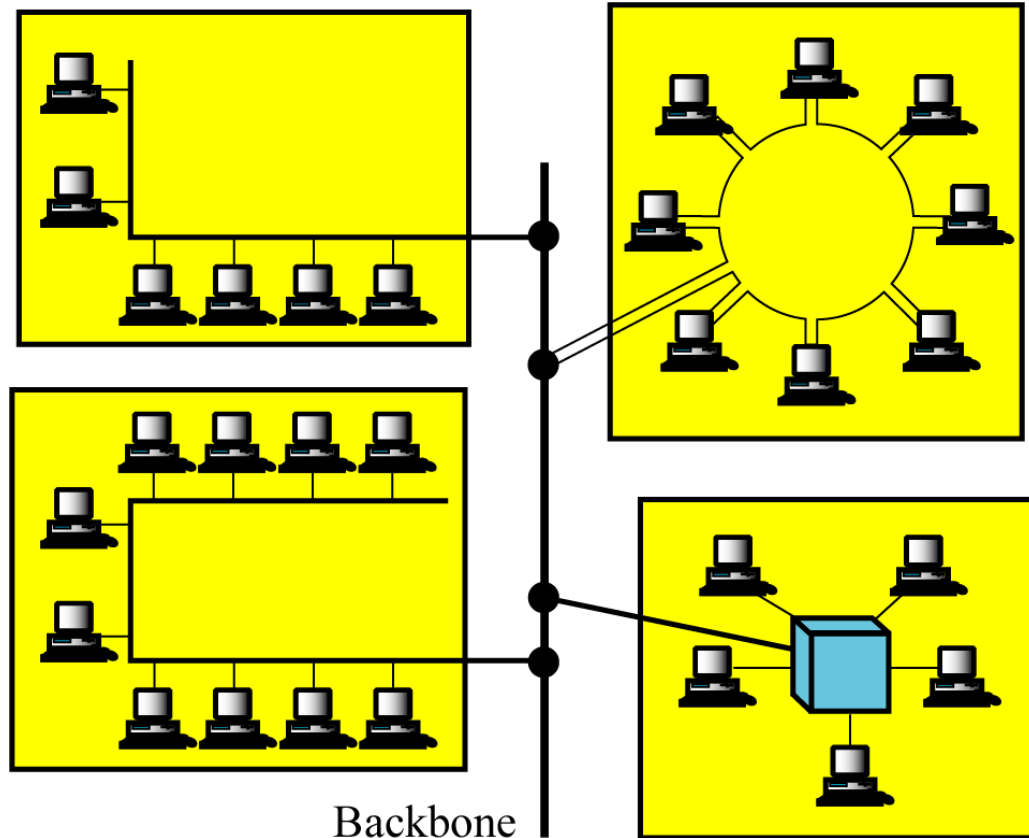
Figure 1.10 *An isolated LAN connecting 12 computers to a hub in a closet*



Categories of Networks (cont'd)



a. Single building LAN

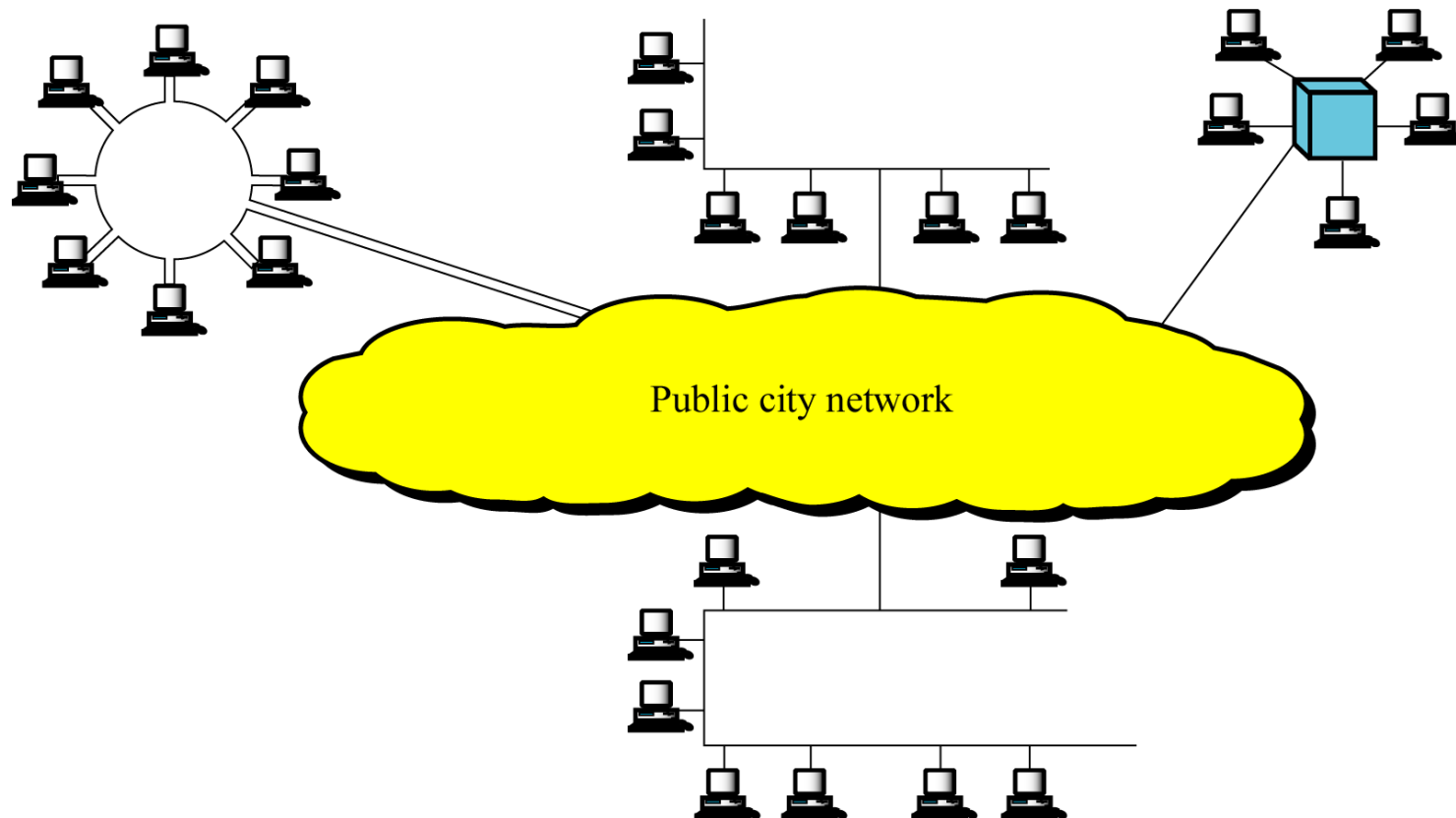


b. Multiple building LAN

Categories of Networks (cont'd)

❑ MAN (Metropolitan Area Networks)

- ◆ is designed to extend over an entire city



Categories of Networks (cont'd)

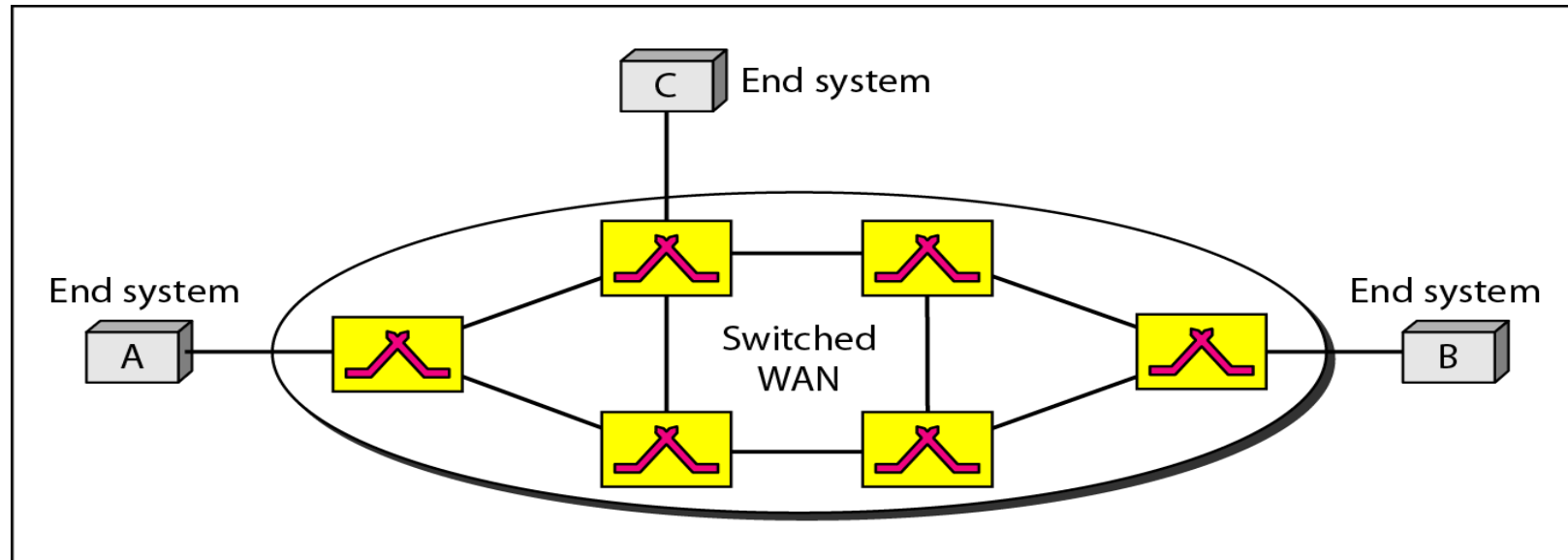
❑ WAN(Wide Area networks)

- ◆ provides long-distance transmission of data, voice, image, and video information over large geographical areas that may comprise a country, a continent, or even the whole world

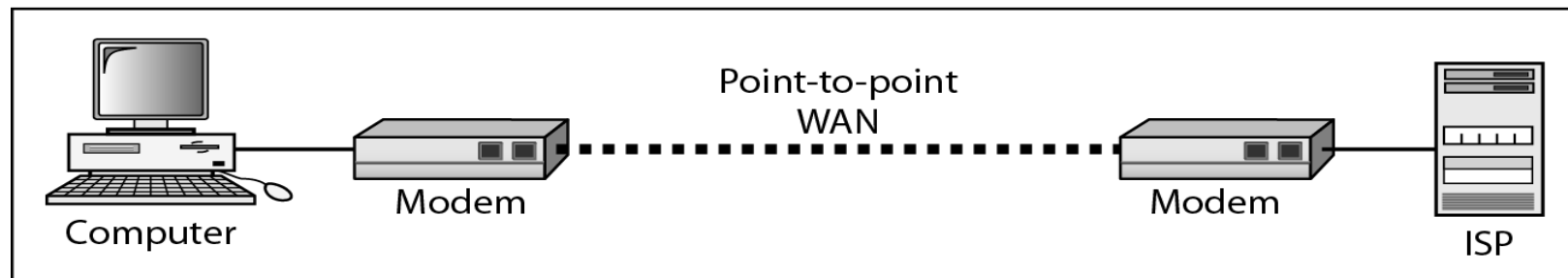


Categories of Networks (cont'd)

Figure 1.11 *WANs: a switched WAN and a point-to-point WAN*



a. Switched WAN

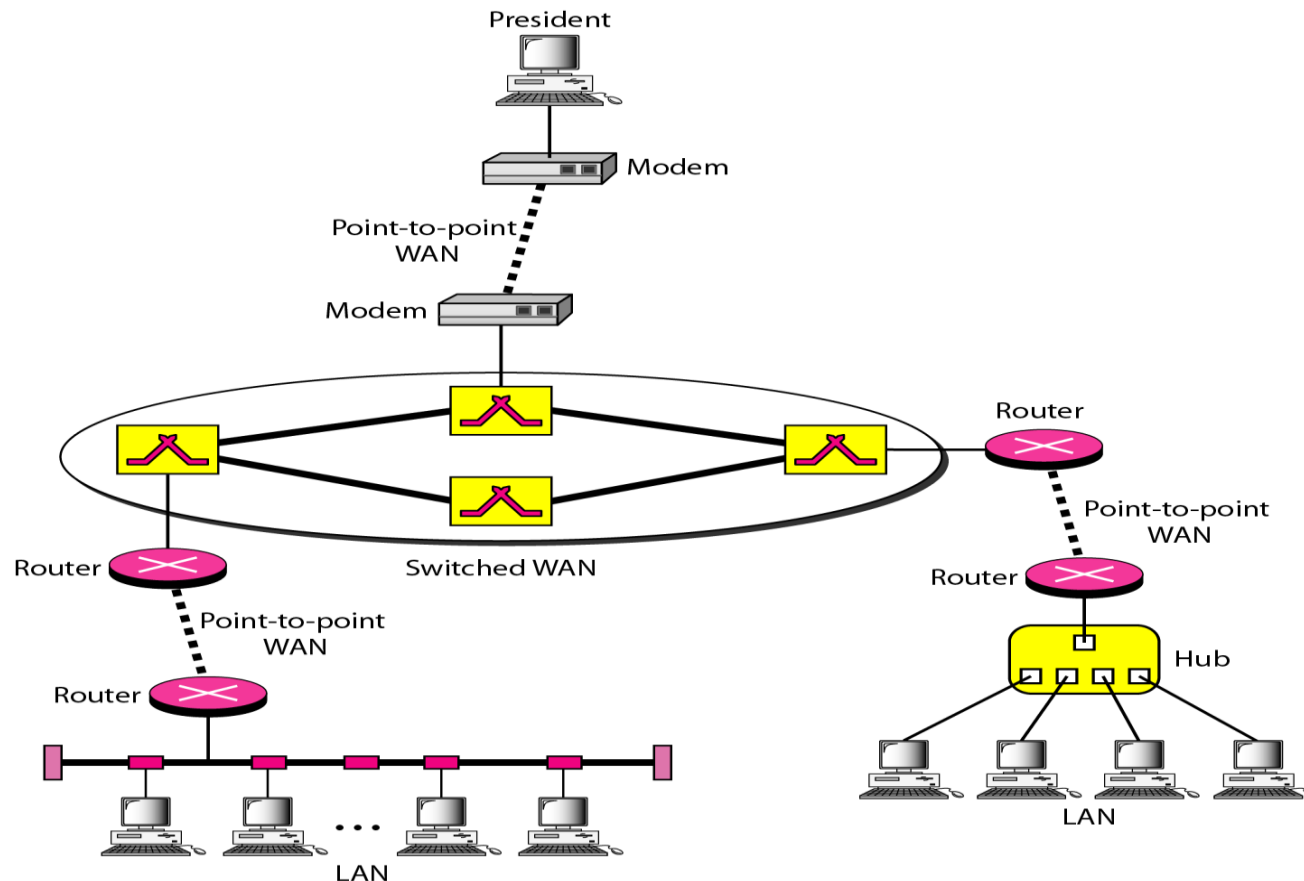


b. Point-to-point WAN

Interconnection of Networks : Internetwork

- Two or more networks are connected, they become an internet.

Figure 1.12 *A heterogeneous network made of four WANs and two LANs*



1.3 THE INTERNET

The **Internet** has revolutionized many aspects of our daily lives. It has affected the way we do business as well as the way we spend our leisure time. The Internet is a communication system that has brought a wealth of information to our fingertips and organized it for our use.

Topics discussed in this section:

A Brief History

The Internet Today (ISPs)



❏ A Brief History

- ❖ Internet (not internet)
- ❖ ARPANET (Advanced Research Projects Agency: ARPA) – mid-1960s
 - Advanced Research Projects Agency (ARPA) in the DOD : presented ideas of ARPANET, 1967
 - Reality : 1969 (UCLA, UCSB, SRI, U of Utah)
- ❖ Birth of Internet : in 1972
 - Vint Cerf and Bob Kahn : Internetting Project
- ❖ TCP/IP : A paper by Cerf and Kahn in 1973
 - In 1983, TCP/IP became to official protocol for the ARPANET
- ❖ MILNET : in 1983
- ❖ CSNET, NSFNET, ANSNET, ...

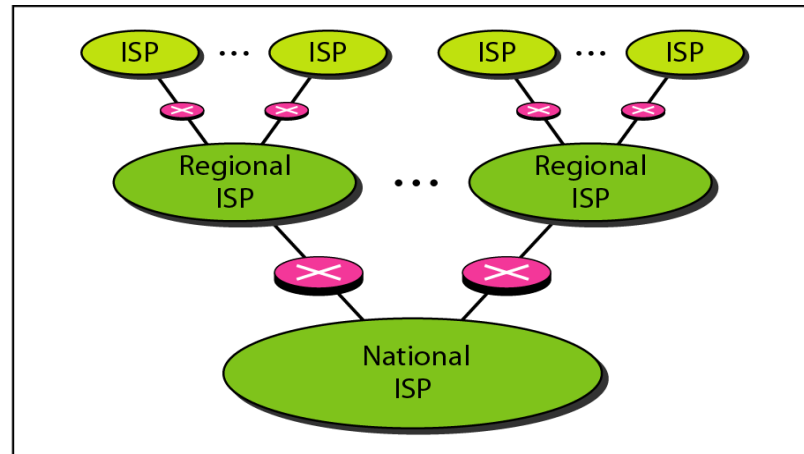
Internet Today

- ❑ **1969.** Four-node ARPANET established.
- ❑ **1970.** ARPA hosts implement NCP.
- ❑ **1973.** Development of TCP/IP suite begins.
- ❑ **1977.** An internet tested using TCP/IP.
- ❑ **1978.** UNIX distributed to academic sites.
- ❑ **1983.** TCP/IP becomes the official protocol
- ❑ **1983.** MILNET was born.
- ❑ **1986.** NSFNET established.
- ❑ **1986,** **IETF First Meeting**
- ❑ **1990.** ARPANET replaced by NSFNET.
- ❑ **1991.** **WWW by CERN**
- ❑ **1995.** NSFNET became a research network.
- ❑ **1995.** **ISPs** started.
- ❑ **2006** **GENI** (Global Environment for Network Innovations) for Future Internet

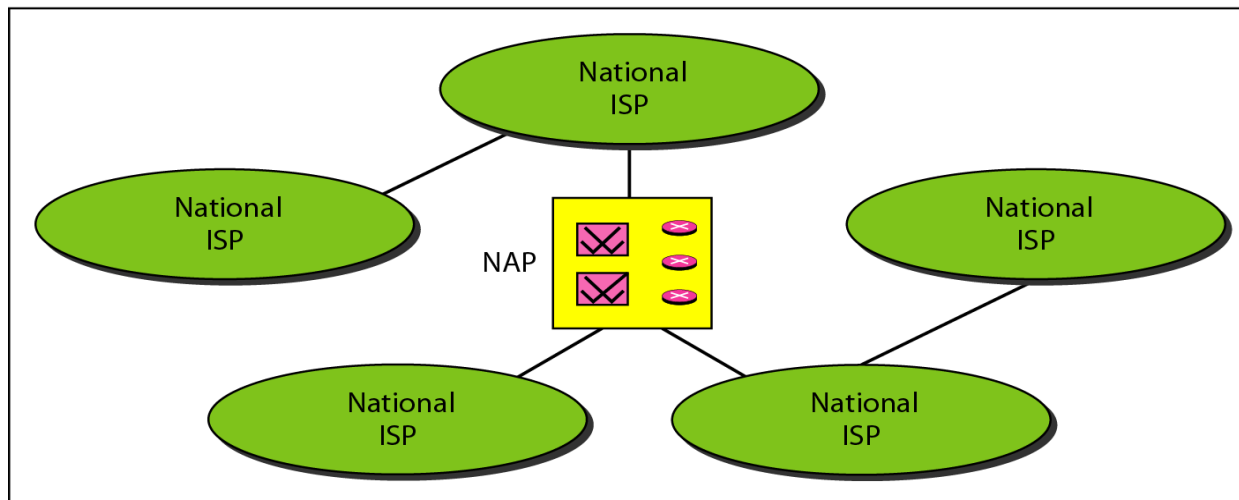


Internet Today

Figure 1.13 *Hierarchical organization of the Internet*



a. Structure of a national ISP



b. Interconnection of national ISPs

1-4 PROTOCOLS AND STANDARDS

In this section, we define two widely used terms: **protocols** and **standards**. First, we define protocol, which is synonymous with rule. Then we discuss standards, which are agreed-upon rules.

Topics discussed in this section:

Protocols

Standards

Standards Organizations

Internet Standards



Protocols and Standard

❑ Communication in computer networks

- ◆ Communication occurs between entities in different systems

❑ Entity

- ◆ is anything capable of sending or receiving information

❑ Protocol

- ◆ is a set of rules that govern data communication



Protocol & Standards (cont'd)

□ Key elements of a Protocol

❖ Syntax

- refers to the structure or format of the data, meaning the order in which they are presented.

❖ Semantics

- refers to the meaning of each section of bits.

❖ Timing

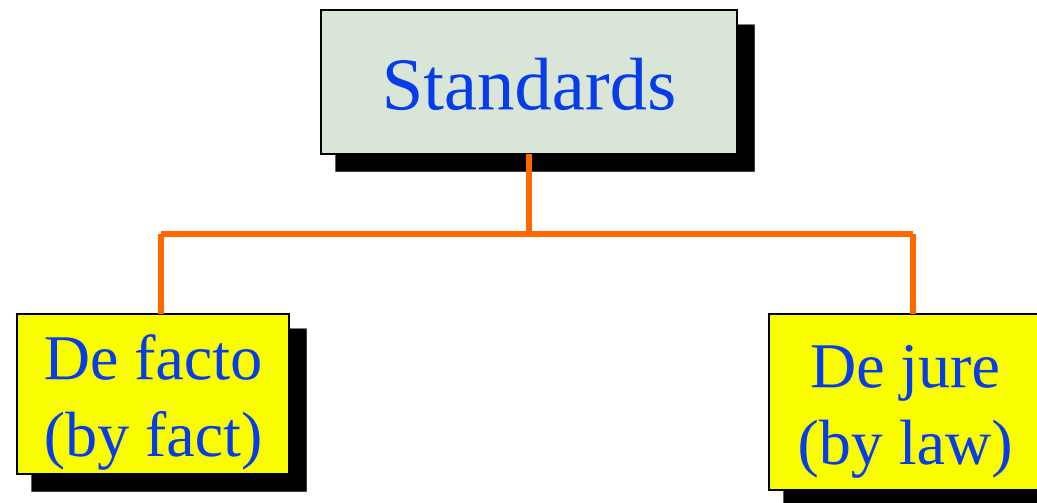
- refers to two characteristics (when data should be sent and how fast it can be sent)



Standards

□ Standards

- ◆ are essential in creating and maintaining an open and competitive market for equipment manufactures and in guaranteeing national and international interoperability of data and communications technology and processes.



Standards Organizations

□ Standard Creation Committees

❖ ISO (The International Standard Organization)

- created in 1947
- Multi national body
- is an organization dedicated to worldwide agreement on international standards in a variety of fields (scientific, technological, economic activity)



Standards Organizations (cont'd)

❖ ITU-T(International Telecommunications Union - Telecommunications Standards Sector)

- is an international standards organization related to the United Nations that develops standards for telecommunications.
- Two popular standards developed by ITU-T are the V series (data transmission over phone lines) and the X series (transmission over public digital networks)



Standards Organizations (cont'd)

❖ ANSI (American National Standard Institute)

- is a nonprofit organization and is the U.S. voting representative to be both the ISO and the ITU-T

◆ IEEE (Institute of Electrical and Electronic Engineers)

- is the largest national professional group involved in developing standards for computing, communication, electrical engineering, and electronics)

◆ EIA (Electronic Industries Association)

- is an association of electronics manufactures in the United States. (EIA-232-D, EIA-530 standards)



Standards Organizations (cont'd)

❑ Forums

- ❖ consist of representatives from corporation that test, evaluate and standardize new technologies.
- ❖ Frame Relay Forum
- ❖ ATM Forum and ATM consortium

❑ Regulatory agencies

- ❖ FCC (Federal Communications Commission)
- ❖ 방송통신위원회
Broadcasting and Communications Commission



Internet Standards

❑ Internet standard by IETF (Internet Engineering Task Force)

- ❖ is a thoroughly tested specification that is useful to and adhered by those who work with the Internet

❑ A specification begins as an Internet draft

- ❖ Working document with no official status and six-month lifetime

❑ RFC (Request for Comment) Recommendation from Internet authorities

Standards (cont'd)

❑ by law standard

~ are those that have been legislated by an officially recognized body.

❑ by fact standard

~ are often established originally by manufactures seeking to define the functionality of a new product or technology.

❖ proprietary (closed)

~ are those originally invented by a commercial organization as a basis for the operation of its products.

❖ nonproprietary (open)

~ are those originally developed by group or committees that have passed them into the public domain.



Standards Organizations (cont'd)

□ Telecommunications Technology Association

한국정보통신기술협회(TTA) : <http://www.tta.or.kr>

◆ ITU related standardization

□ Korean Agency for Technology and Standards

한국기술표준원[KATS] : <http://www.kats.go.kr>

◆ ISO related standardization



Summary (1)

- ❑ **Data communications** are the transfer of data from one device to another via some form of transmission medium.
- ❑ A data communications system must transmit data to the correct destination in an accurate and timely manner.
- ❑ The five components that make up a data communications system are the message, sender, receiver, medium, and protocol.
- ❑ Text, numbers, images, audio, and video are different forms of information.
- ❑ Data flow between two devices can occur in one of three ways: **simplex, half-duplex, or full-duplex.**
- ❑ A **network** is **a set of communication devices connected by media links.**
- ❑ In a point-to-point connection, two and only two devices are connected by a dedicated link. In a multipoint connection, three or more devices share a link.

Summary (2)

- ❑ **Topology** refers to the physical or logical arrangement of a network. Devices may be arranged in **a mesh, star, bus, or ring** topology.
- ❑ A network can be categorized as a local area network or a wide area network.
- ❑ A LAN is a data communication system within a building, plant, or campus, or between nearby buildings.
- ❑ A WAN is a data communication system spanning state, countries, or the whole world.
- ❑ An internet is a network of networks.
- ❑ **The Internet is a collection of many separate networks.**
- ❑ There are local, regional, national, and international Internet service providers.

Summary (3)

- ❑ A protocol is a set of rules that govern data communication; the key elements of a protocol are syntax, semantics, and timing.
- ❑ Standards are necessary to ensure that products from different manufacturers can work together as expected.
- ❑ The ISO, ITU-T, ANSI, IEEE, and EIA are some of the organizations involved in standards creation.
- ❑ Forums are special-interest groups that quickly evaluate and standardize new technologies.
- ❑ A Request for Comment (RFC) is an idea or concept that is a precursor to an Internet standard.

Q & A

